

TRAINING PROPOSAL

TRACE-RX

Transformation-Reliability-Aware AI Image Detection

Do not force every forensic feature to vote on every image. Measure whether the signal survived, model correlated evidence, and abstain when the endpoint no longer contains enough information.

Multiple evidence families

PE provenance semantics, DINOv3 structure, native residuals, frequency, phase and authentic manifolds.

Controlled interventions

Mild, content-preserving perturbations measure whether fragile forensic evidence remains observable.

Reliable fusion

Availability, uncertainty and covariance determine how strongly each expert can speak.

Operational decision

Return real, AI-generated or insufficient evidence with calibrated confidence.

Execution target: two DGX Sparks, 24 hours, open models below 2B parameters, and a lineage-safe 160,000-record dataset.

01 / PROBLEM AND BACKGROUND

Why robust AIGC detection is difficult

Platform problem

Highly realistic synthetic images can support misinformation, impersonation and fraud. The practical target is not a pristine generator output: it is the endpoint after compression, cropping, filters, reposting, screenshots or recapture.

Competition scope

- Binary image-level AIGC detection with confidence scores.
- Robustness to JPEG, blur, resize, noise, color jitter and crop.
- Strong generalisation and careful authentic-image false-positive control.
- Convincing proof of concept under limited compute; model below 2B parameters.

Statistical target

Let $Y=1$ denote a fully synthetic origin and $Y=0$ an authentic origin. A post-processing journey T changes pixels but not origin: $Y(T(x))=Y(x)$. The detector estimates $p(Y=1 \mid T(x))$ while maximising sensitivity at a fixed, low authentic-image false-positive rate.

What the literature establishes

No universal fingerprint

Generator artifacts vary by model family and decoder generation. A detector that wins on one benchmark may fail on current generators.

Signals fail differently

Semantic evidence can survive compression but carries content bias. Fine residual and frequency evidence can be discriminative yet fragile.

Missing evidence is asymmetric

Positive camera evidence supports authenticity. Its absence does not prove synthesis because ordinary editing can erase it.

Transformations have order

Resize then JPEG is not equivalent to JPEG then resize. Repeating a lossy operation can produce saturation rather than a proportional change.

02 / DETECTION FOUNDATIONS

What information can distinguish synthetic and authentic origin?

Evidence family	Measured phenomenon	Representative statistic	Principal failure
Semantic and physical consistency	Objects, text, lighting, reflections and generator-associated visual style	Learned embedding $\phi(x)$; class logit $w^T \phi(x) + b$	Content and style shortcuts; generators improve
Local RGB and neighbour relations	Decoder texture, upsampling and short-range pixel dependencies	$C(\delta) = E[(x_{p-\mu})(x_{p+\delta}-\mu)]$	Resize, blur and denoise alter local dependence
Residual and camera evidence	Sensor noise, demosaicing, sharpening and synthesis residuals	$r = x - G_{\sigma} * x$; residual energy and correlation	Compression and editing erase genuine camera evidence
Frequency magnitude	Periodicity, spectral slope and decoder upsampling artifacts	$A(u,v) = \log(1 + \text{IFFT}(x)_{\{u,v\}})$	Easy to erase or imitate; JPEG shortcuts
Frequency phase	Spatial arrangement encoded separately from magnitude	$\theta(u,v) = \text{atan2}(\text{Im FFT}, \text{Re FFT})$	Crop and geometric resampling change phase
Authentic acquisition manifold	Positive similarity to camera, screenshot, scan, artwork or conventional-CGI subtypes	$d_c(x) = \ \phi(x) - \mu_{\text{c}}\ ^2$	Absence of authentic evidence is not proof of AI
Provenance and metadata	C2PA assertions, watermarks and file/container history	Verified positive credential or watermark	May be removed or absent; absence is non-evidence

Why multiple experts

The evidence families are conditionally informative under different transformations. A semantic feature may survive JPEG while a residual feature disappears.

Why missing evidence is asymmetric

A verified sensor or provenance trace supports authenticity. Its absence is compatible with both synthesis and ordinary redistribution.

Why fusion needs reliability

Concatenation assumes every feature remains trustworthy. TRACE-RX instead estimates signal availability and correlation before combining logits.

Design principle

The model should learn evidence of origin, not a particular file format, image subject or generator name. Every label-correlated acquisition choice is therefore treated as a potential confound and tested with source-disjoint splits and metadata-only baselines.

03 / GLOBAL REPRESENTATION EXPERTS

Three backbones, two representation paradigms

PE-Core-L, DINOv3-L and DINOv2-L are the only pretrained backbones in the trial. PE represents images in a language-aligned space; DINOv3 and DINOv2 provide self-supervised dense visual structure. One primary encoder is selected before any optional two-backbone fusion is considered.

Backbone	Scale	Why consider it	Main risk	Decision
PE-Core-L/14	320M vision	Language-aligned provenance semantics; Apache 2.0; supports semantic prototype calibration	Best direct AIGC evidence is at 1.9B G scale; L-scale gain must be measured	Provenance candidate
DINOv3-L/16	300M	Strong dense image features; current dominant frozen vision-only AIGC baseline	Custom gated license; largest published AIGC baseline used prohibited 6.7B model	Structure candidate
DINOv2-L/14	300M	Apache 2.0; mature and reproducible dense features	Older pretraining and weaker current foundation-model baseline	Reproducible control

PE semantic prototypes

Normalize image embedding e and text prototypes t_c ; compute $z_c = (e^T t_c) / \tau$. Calibrate prototypes without forcing the backbone to relearn vision.

Three-backbone trial

Freeze PE-L, DINOv3-L and DINOv2-L; cache features; use the same head and schedule. Select one primary global encoder by low-FPR and worst-group metrics.

One optional fusion test

Add the runner-up once after selecting the best encoder. Keep two global backbones only when their error sets are measurably complementary.

Language-aligned provenance

For normalized PE image embedding e and class-text prototype t_c , $z_c = (e^T t_c) / \tau$. Prototype calibration moves class directions using training data while keeping the visual encoder stable. Authentic subtype prompts reduce the risk that only camera photographs define 'real'.

Self-supervised structure

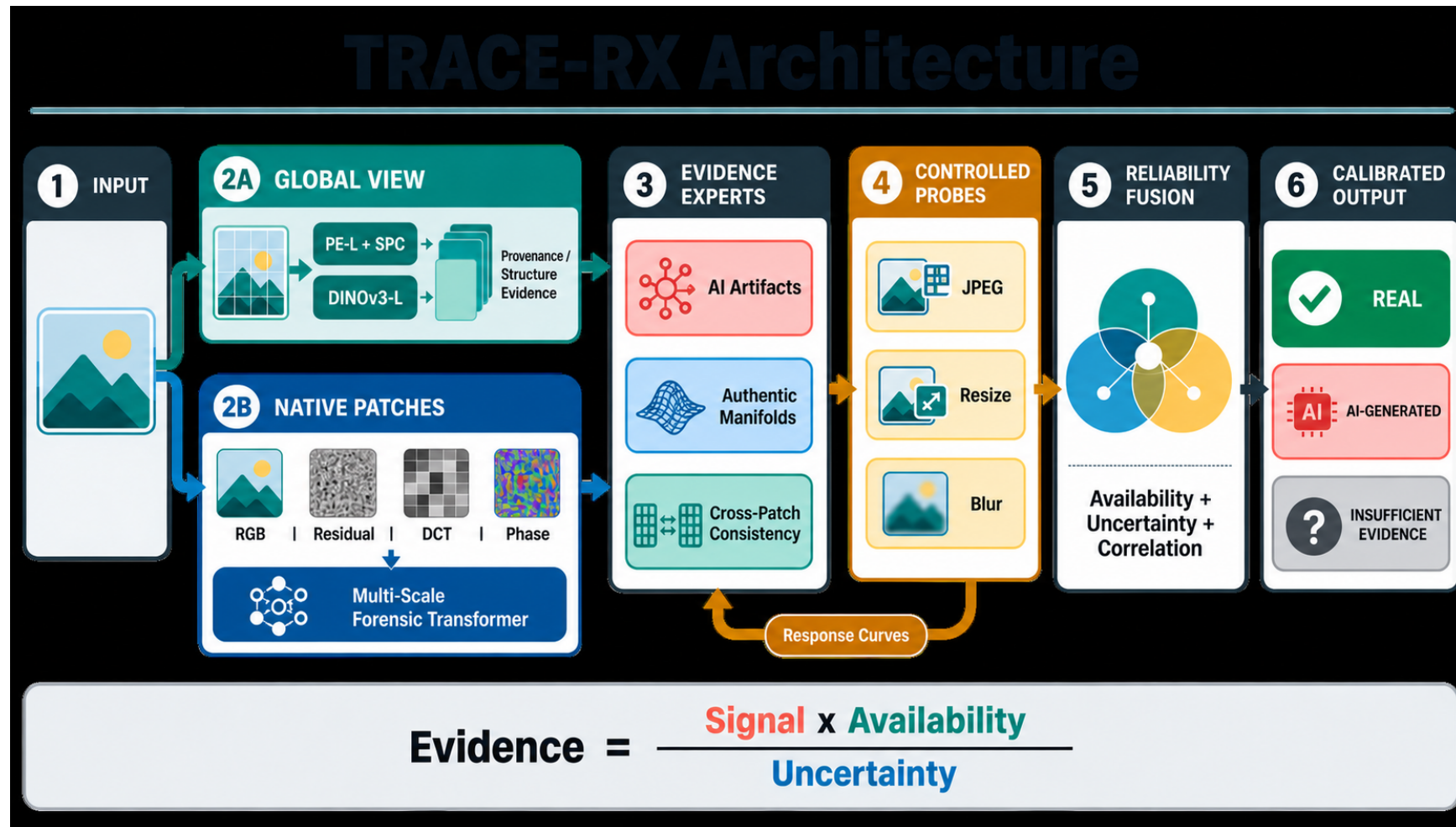
For DINO patch tokens $h_1, \dots, h_n$, form $h_{\text{pool}} = \text{mean}_i h_i$ together with token mean/variance. A shared linear head $z = w^T h_{\text{pool}} + b$ tests whether the representation separates origin without architecture-specific tuning.

Evidence note: the August 2026 PE-SPC preprint reports PE-SPC matching or exceeding a DINOv3 linear probe across cross-generator, in-the-wild, recent-generator and blur/JPEG groups. Its headline models were PE-G/14 at 1.9B and DINOv3-7B at 6.7B; those results do not establish the winner at our 300M scale, so direct ablation remains mandatory.

04 / SYSTEM ARCHITECTURE

TRACE-RX: evidence experts with reliability-aware fusion

TRACE-RX combines one selected global representation backbone with a native-resolution forensic expert, authentic-manifold evidence and controlled intervention responses. JPEG, resize and blur in the diagram are illustrative intervention operators; the final intervention set is selected empirically.



Core evidence score: effective evidence = signed expert evidence x availability / uncertainty. Intervention responses are fusion features, not independent votes. A constrained covariance layer reduces double-counting when experts react to the same artifact.

05 / NATIVE FORENSIC EXPERT

Preserve local evidence that global resizing can destroy

A single 224-pixel global resize can erase the exact microtexture, sampling and residual patterns that image forensics needs. The forensic expert therefore analyses native-resolution patches selected from high texture, low texture, edges, faces/text and uniform spatial coverage.

Input representation	Construction	What the branch can learn
RGB	Native 192-256 pixel patches without global downsampling	Texture, edges, text rendering and local physical inconsistencies
High-pass residual	$r_{\sigma}(x)=x-G_{\sigma}x$, with σ fixed or lightly learned	Sensor/demosaicing traces, sharpening, decoder and denoising residuals
Block DCT	$D_{\{u,v\}}=\sum x_{\{i,j\}} \cos_u(i) \cos_v(j)$ over local blocks	Quantisation, periodicity, spectral slope and block-grid artifacts
Fourier phase	$\theta_{\{u,v\}}=\text{atan2}(\text{Im FFT}(x), \text{Re FFT}(x))$	Spatial organisation that magnitude-only detectors discard
Neighbour relations	$C_{\delta}=E[(x_p-\mu)(x_{p+\delta}-\mu)]$ for short offsets δ	Upsampling, interpolation and decoder-specific local dependence

Patch encoding
Each modality is tokenised at multiple scales. A 40-60M transformer shares spatial locations across modalities while retaining modality embeddings.

Patch aggregation
For patch logits z_i , retain mean, variance, upper quantile and attention-weighted score. High disagreement is evidence of locality or uncertainty.

Forensic uncertainty
Predict log variance $v(x)$ with the logit. The fusion system can discount a confident-looking score when the expert's learned variance is high.

Cross-patch consistency

If z_i is the AI logit for patch i , define $V_{\text{patch}}(x)=\text{Var}_i[z_i]$. The model receives both mean evidence and V_{patch} . This provides crop robustness without deliberately discarding 20% of the image as an inference-time intervention.

06 / DATA AND COMPUTE PROTOCOL

Use 95% of independent masters for model fitting

A. Lineage-safe allocation

- 80,000 independent masters: 40,000 authentic and 40,000 fully AI-generated.
- One journey endpoint per master creates 160,000 records; derivatives are not independent observations.
- 76,000 masters (152,000 records) are assigned to expert or fusion fitting.
- The locked set contains exactly 2,000 image records from 1,000 masters: 500 authentic and 500 synthetic lineages.
- A master, derivative, prompt group and duplicate component always remain in one partition.
- The locked records are the clean and journey endpoint for each test master; wider transform experiments use development.

Partition	Masters	Records	Role
Expert train	72,000	144,000	Fit experts
Fusion train	4,000	8,000	Fit fusion/objectives
Development	2,000	4,000	Select methods
Calibration	1,000	2,000	Thresholds only
Locked test	1,000	2,000	Exactly 2,000 images; open once

Test-set precision caveat

With 500 authentic test masters, one master-level false positive changes FPR by 0.2 percentage point. Report exact binomial or Wilson intervals and avoid production-level claims; this cohort provides fast independent hackathon validation.

B. Independent work queues

Spark A - global and modelling

Cache PE-L, DINOv3-L and DINOv2-L features; select one primary backbone; fit fusion, robust objectives and calibration.

Spark B - forensic and transforms

Train the native-patch forensic tower; compute candidate interventions, official transform sweeps, interactions, order effects and repeats.

Shared contract

Identical immutable manifests; separate output directories; exchange compact feature tables rather than transformed image corpora.

No risky cluster setup

Use multi-node training only if networking is already configured and passes a 15-minute NCCL test. Otherwise keep both Sparks productive independently.

07 / CONTROLLED INTERVENTIONS

Estimate whether forensic evidence remains observable

A controlled intervention is a mild, deterministic transformation applied temporarily at inference. It is a stress test, not a new training label and not a vote. The model compares an expert's representation before and after intervention to estimate local sensitivity at the observed endpoint.

Candidate	What it stresses	Interpretation risk
JPEG q95	Quantisation, grid and residual evidence	Small response may mean prior saturation
Blur sigma 0.5	High-frequency texture and noise	Overlaps with resize and denoise
Resize 0.95 then restore	Sampling grid and neighbour relations	Interpolation choice can dominate
Saturation 0.95	Colour dependence and filter shortcuts	May alter semantic appearance
Mild denoise	Residual and noise evidence	Can erase genuine sensor evidence
Cross-patch consistency	Local evidence and crop dependence	Content heterogeneity creates variance

Why interventions help

An expert may output a strong logit from evidence that disappears under an imperceptible perturbation. The response exposes this fragility to fusion.

Why several operators

Compression, low-pass filtering, resampling, colour change and patch selection disturb different statistics. Complementarity matters more than count.

How operators are chosen

Screen all candidates on development; keep an operator only for measurable low-FPR, worst-group or calibration gain without authentic-FPR harm.

Response representation

For expert feature h and logit z , define $r_p(x) = [|z(x) - z(T_p(x))|, 1 - \cos(h(x), h(T_p(x))), \|h(x) - h(T_p(x))\|_2 / (\|h(x)\|_2 + \epsilon)]$. A small response is interpreted jointly with original evidence strength and estimated image quality: it may indicate stable evidence or an already-saturated image.

Intervention-set utility: $U(P) = \Delta \text{TPR@1\%FPR} + \alpha \Delta \text{worst-group AUC} - \lambda \text{latency}(P) - \mu \text{instability}(P)$. Greedy forward selection stops when marginal held-out utility becomes negligible.

08 / RELIABILITY-AWARE FUSION

Combine evidence according to availability, uncertainty and correlation

Suppose K experts emit aligned AI logits $z=(z_1, \dots, z_K)$. A conventional concatenation head can let a fragile expert dominate after its signal is erased. TRACE-RX separately estimates whether each expert can speak and how uncertain its score is.

Quantity	Definition	Role
Evidence z_k	Signed AI logit from expert k	Direction and strength of origin evidence
Availability a_k	$a_k = \text{sigmoid}(g_k(q(x), r(x)))$	Whether quality and intervention responses indicate that signal k remains observable
Variance s_k^2	Positive heteroscedastic uncertainty predicted by expert k	Downweight examples for which the expert is intrinsically uncertain
Covariance Σ	Low-rank covariance of expert residual errors	Avoid counting correlated texture/frequency evidence several times

Generalised reliability weighting

Let a be the availability vector. Define $w = \Sigma^{-1}a / [a^T \Sigma^{-1}a + \epsilon]$, then $z_{\text{fused}} = w^T z$ and $p_{\text{AI}} = \text{sigmoid}(z_{\text{fused}}/T)$, where T is fitted on calibration.

- Diagonal Σ reduces to availability- and inverse-variance weighting.
- Low-rank off-diagonal terms model correlated expert mistakes.
- Constrain weights or use logistic stacking if covariance fitting is unstable.

Abstention and guardrails

- Abstain when predictive entropy is high, expert disagreement is high or total available evidence is low.
- Treat verified C2PA or watermark evidence as an independent positive provenance channel.
- Never infer AI solely from absent EXIF, PRNU, watermark or provenance credentials.
- Calibrate real/AI thresholds and the insufficient-evidence region separately.

09 / ARCHITECTURE EXPERIMENTS

Identify the smallest evidence system that generalises

ID	Global evidence	Forensic	Authentic prototypes	Interventions	Purpose
B0	PE-L	-	-	-	Backbone trial 1
B1	DINOv3-L	-	-	-	Backbone trial 2
B2	DINOv2-L	-	-	-	Backbone trial 3
A0	Best of B0-B2	-	-	-	Primary global baseline
A1	Best	Yes	-	-	Add local forensic evidence
A2	Best	Yes	Yes	-	Add positive-real evidence
A3	Best	Yes	Yes	Selected	Full reliability system
A4 optional	Best + runner-up	Yes	Yes	Selected	Single complementarity check

Selection score

$S = 0.35 \text{ TPR@1\%FPR} + 0.25 \text{ worst-group AUC} + 0.20 \text{ unseen-generator AUC} + 0.10(1 - \text{worst-real FPR}) + 0.10(1 - \text{ECE})$.

- Run exactly three pretrained backbones under one frozen-feature protocol.
- Use one seed for component screening; rerun only the finalist.
- Compare expert error correlation, not only fused AUC.
- Test a second global backbone only once, after choosing the primary encoder.

Removal rules

- Drop a branch with less than 0.5 point fused gain.
- Drop a branch whose gain disappears at low FPR or on unseen generators.
- Drop any component that raises worst authentic-subtype FPR materially.
- Fallback: if the custom forensic tower is unstable by hour 5, use a proven lightweight RGB/residual backbone.

10 / TRAINING OBJECTIVES

Optimise origin accuracy, worst-group risk and low-FPR operation

The objectives are evaluated as separate hypotheses and then, if useful, staged. This prevents several losses from simultaneously pulling the representation in incompatible directions.

Variant	Definition	Question
O0 Balanced BCE	Master-, class-, subtype- and generator-balanced binary cross-entropy	Strong reference objective
O1 BCE + CVaR	Add loss over the worst 20% of hierarchical groups	Does worst-group robustness improve?
O2 BCE + pAUC	Rank AI positives above hard authentic negatives near the low-FPR boundary	Does authentic false-positive control improve?
O3 Staged	BCE convergence -> CVaR fine-tune -> short pAUC stage -> separate calibration	Can sequencing avoid objective conflict?

Balanced BCE

$L_{\text{BCE}} = -E[w_g \{y \log p + (1-y) \log(1-p)\}]$. Group weights w_g prevent dominant sources or generators from controlling the gradient.

Worst-group CVaR

$L_{\text{CVaR}} = \min_{\eta} [\eta + E(L_g - \eta)_+ / (1-\alpha)]$, with $\alpha=0.8$. It emphasizes the worst 20% of group losses.

Low-FPR ranking

$L_{\text{pAUC}} = \text{mean softplus}(m - z_{\text{pos}} + z_{\text{hard-real}})$, where hard-real examples lie near the target false-positive boundary.

Conflict diagnostic and operational sequence

For losses L_i and L_j , monitor $C_{ij} = (\text{grad } L_i \cdot \text{grad } L_j) / (\| \text{grad } L_i \| \| \text{grad } L_j \|)$ on 256 batches. Persistent $C_{ij} < 0$ with no held-out gain indicates objective interference.

- Train expensive experts once on the 144,000 expert-training records.
- Cache expert outputs; run architecture and objective sweeps as inexpensive fusion experiments.
- Fit fusion on the separate 8,000-record fusion partition; never fit it on development.
- Fit temperature and abstention thresholds only on calibration after architecture and objective choices freeze.

11 / TRANSFORMATION EXPERIMENTS

Measure signal survival, interaction, order and saturation

Experiment	Design	Report artifact
Official single transforms	JPEG 90/70/50/30; blur 0.5/1/2; resize 0.5/0.25; noise 0.02/0.05/0.10; colour +/-20%; crop 80%	Branch x severity survival heatmap
Interactions	Eight-condition fractional-factorial design on 2,000 development masters	Pairwise interaction matrix
Order	Resize/JPEG, blur/JPEG and crop/resize in both directions	Order-effect comparison
Repeated operation	JPEG -> JPEG; resize -> resize; blur -> blur with every stage scored	Saturation curves
Generalisation	Unseen generator, unseen authentic source and journey-held-out groups	Low-FPR and worst-group table

For branch k under condition t, standardised class separation is $d'(k,t)=[\mu_{AI}(k,t)-\mu_{real}(k,t)] / \sqrt{[\text{var}_{AI}(k,t)+\text{var}_{real}(k,t)]/2}$. This measures discrimination independently of a fixed classification threshold.

Signal survival
For branch k and transform t, $\rho(k,t) = d'(k,t) / [d'(k,clean) + \epsilon]$. Negative survival means the signal has become misleading.

Interaction
 $I(a,b) = d'(a \text{ after } b) - d'(a) - d'(b) + d'(clean)$. A negative value indicates super-additive damage.

Order and saturation
Compare a after b with b after a; score every repeated-operation stage so a small second change is not mistaken for preserved evidence.

Uncertainty and error analysis

Cluster-bootstrap by master lineage. Each high-confidence error records source/generator, authentic subtype, complete transform lineage, expert scores, selected-intervention responses, uncertainty and the exact stage where the decision changed.

12 / EVALUATION MATHEMATICS

Report discrimination, low-FPR utility, calibration and abstention

Metric	Definition	Interpretation
ROC-AUC	$P[s(x_{AI}) > s(x_{real})]$ with ties assigned one half	Threshold-free ranking; can hide low-FPR failure
TPR at fixed threshold	$TPR(\tau) = P[s \geq \tau \mid Y=1]$	Synthetic-image sensitivity at an operational threshold
FPR at fixed threshold	$FPR(\tau) = P[s \geq \tau \mid Y=0]$	Authentic-image harm; primary guardrail
Brier score	$Brier = N^{-1} \sum_i (p_i - y_i)^2$	Overall probability accuracy
Expected calibration error	$ECE = \sum_b (n_b/N) accuracy_b - confidence_b $	Whether confidence matches empirical frequency
Risk at coverage c	$R(c) = \text{error rate among the } c \text{ fraction with lowest uncertainty}$	Value of abstaining on insufficient evidence
Worst-group metric	$\min_g \text{metric}_g \text{ over predeclared generator/source/journey groups}$	Prevents a good mean from hiding systematic collapse

Threshold protocol

- Choose τ from authentic calibration scores, never from the locked test.
- For a nominal 1% calibration FPR, use the empirical 99th percentile or a conservative quantile bound.
- On locked testing, report the realised $FPR(\tau)$ and $TPR(\tau)$; never retune τ to force 1% test FPR.

Uncertainty protocol

- Resample master lineages so clean and journey endpoints remain paired.
- Use Wilson or exact binomial intervals for FPR/TPR with the small locked cohort.
- Report per-seed metrics for finalists; do not treat multiple seeds as independent images.

13 / 24-HOUR EXECUTION

Two work queues, four decision gates, one locked test

Hours	Spark A - global/modelling	Spark B - forensic/transforms	Gate / output
0-1	Manifest, split and decoder checks	Transform parity and 200-image smoke test	Immutable protocol
1-5	Cache PE, DINOv3 and DINOv2 features	Train forensic tower; prepare dev interventions	Fallback tower if unstable
5-9	B0-B2 backbone trial; A0-A3 components	Intervention and single-transform sweep	Primary backbone; selected interventions
9-14	O0 BCE and O1 CVaR	Selected responses; O2 pAUC and O3 staged	Freeze objective
14-18	Train finalist fusion	Interactions, order and repeated operations	Freeze architecture and weights
18-20	Calibration and abstention	Complete transformed evaluation	Freeze thresholds
20-21	Open 2,000-image locked metrics once	Locked forensic inference	No further model changes
21-24	Tables, calibration and narrative	Heatmaps, curves and error cards	Reproducible report package

Mandatory evidence

- Three-backbone trial plus global/forensic expert-error correlation.
- No-intervention versus selected-intervention marginal gains.
- BCE versus CVaR, with pAUC only if stable.
- Official single transforms and repeated JPEG/resize/blur.
- Calibration, master-clustered intervals and locked-test discipline.

Kill order if late

- Second forensic seed.
- Denoise and saturation interventions.
- Partial-AUC if unstable.
- Lower-priority interaction pairs.
- Full locked transformation matrix; preserve final clean and journey evaluation.

The 2,000-image locked set supports rapid independent validation, not production certification. All selection, intervention screening and objective tuning finish before its labels are opened.

14 / ERROR ANALYSIS

Turn failures into explanations and the next data decision

Error analysis operates on master lineages, not individual transformed records. Every false positive, false negative and abstention is joined to source, generator, authentic subtype, content stratum and complete transformation history.

Failure class	Diagnostic evidence	Likely response
Source or format shortcut	Metadata-only model predicts label; errors concentrate by codec/source	Rebalance acquisition; canonicalise model inputs while preserving masters
Transformation erasure	Branch d' collapses at a specific stage; availability remains high	Correct availability targets; add matched transformed training views
Semantic confusion	Global expert fails on a content subtype while forensic expert remains stable	Collect matched real/AI content; revise prototype prompts
Authentic-manifold mismatch	Real screenshots, art or CGI are far from every authentic prototype	Add subtype prototypes and hard authentic examples
Generator novelty	Unseen generator errors with stable transformation response	Expand generator diversity; favour generator-invariant evidence
Calibration failure	Ranking remains good but confidence is wrong after degradation	Quality-aware temperature scaling or wider abstention region

Statistical decomposition

Model error probability with logit

$P(E=1) = \beta_0 + u_{\text{source}} + v_{\text{generator}} + w_{\text{transform}} + r_{\text{content}} + (v \times w)_{\text{interaction}}$.

Large random-effect estimates identify systematic groups rather than isolated examples.

- Use master-clustered bootstrap intervals for all metrics.
- Use paired score deltas because clean and transformed views share a master.
- Report Wilson or exact binomial intervals for low-count test FPR.

Required report artifacts

- Three-backbone table and architecture component ladder.
- Signal-survival heatmap and interaction/order matrices.
- Intervention marginal-utility curve and response examples.
- Objective Pareto plot, calibration curve and risk-coverage plot.
- Representative error cards with expert and stage-wise scores.

Selected foundations: Wang et al., CVPR 2020; Ojha et al., CVPR 2023; GenImage, NeurIPS 2023; DINOv2, 2023; DINOv3, 2025; Perception Encoder, NeurIPS 2025; PE-SPC, arXiv 2608.04935; Dual Data Alignment, NeurIPS 2025; RRDataset, ICCV 2025; AIGIBench, NeurIPS 2025. Organiser validation remains sealed from training and selection.